

## The Role of Artificial Intelligence in Cardiology

Ibrahim Elhaddar<sup>1,2</sup>  

### Abstract

Artificial intelligence (AI) is revolutionizing cardiovascular medicine by significantly enhancing diagnostic precision and predictive capabilities. In electrocardiography (ECG), AI demonstrates superior sensitivity and accuracy compared to traditional methods, efficiently detecting conditions such as atrial fibrillation, subtle ST-segment changes, QT prolongation, and even asymptomatic left ventricular dysfunction. Recent studies underscore AI's potential in identifying rhythm abnormalities through consumer-grade devices, enabling real-time monitoring and early intervention. However, limitations persist, notably the reliance on retrospective data and limited follow-up periods.

Cardiovascular imaging, including echocardiography, coronary CT angiography, and cardiac MRI, also benefits substantially from AI applications. AI systems effectively interpret echocardiograms with comparable accuracy to experienced cardiologists, significantly reducing analysis time. Similarly, coronary CT angiography enhanced by AI demonstrates high sensitivity in identifying coronary artery disease. AI-driven cardiac MRI analysis accelerates image processing from minutes to seconds, maintaining diagnostic accuracy.

In cardiovascular risk prediction, AI-driven models have consistently outperformed traditional risk assessment tools. AI achieves higher accuracy in predicting heart failure hospitalizations and post-myocardial infarction survival by integrating multifaceted patient data, though current evidence largely stems from retrospective analyses predominantly involving limited demographic groups.

In conclusion, AI holds considerable promise for improving cardiovascular diagnosis and personalized risk prediction. Future clinical integration necessitates comprehensive prospective studies to confirm reliability and address ethical considerations, particularly regarding patient data privacy.

### Introduction

The evidence provided is based on a focused review of selected studies published between 2018 and 2023 that checked the role of AI in cardiology diagnosis. An emphasis was placed on studies with clinical relevance, and particular attention was given to how these data can be applied in the real world. The review specifically highlights advancements in ECG interpretation, cardiovascular imaging modalities, and risk prediction models, addressing both the strengths and current limitations of AI applications to support clinical decision-making and optimize patient outcomes in cardiology practice.

### AI In ECG Analysis

Electrocardiography (ECG) has always been an essential diagnostic tool when it comes to overt heart conditions, but interpreting subtle ECG changes is not always easy. The growing use of AI offers the potential to make improvements in both the accuracy and speed of ECG analysis.<sup>1</sup> AI shows the capability to identify conditions such as atrial fibrillation (AF), changes in the ST-segment and QT prolongation, and often with improved sensitivity even when compared to human interpretation.<sup>1,2</sup> AI is now also being added to everyday consumer products such as smartwatches, allowing single-lead ECG moni-

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**1** Clinical Tutor, Libyan International University, Benghazi, Libya

**2** Master of Science Student, Department of Medicine, University of Edinburgh, United Kingdom

**Corresponding Author:**

[i.h.m.elhaddar@sms.ed.ac.uk](mailto:i.h.m.elhaddar@sms.ed.ac.uk)

toring. Further advancements will likely allow real-time detection of heart rhythm abnormalities, improving early diagnosis.<sup>1</sup> The involvement of AI in ECG interpretation can go beyond just rhythm analysis, with AI showing promise in detecting subtle cardiac dysfunction.<sup>2</sup> One such example is a deep learning model that, when applied to a 12-lead ECG was able to screen for asymptomatic left ventricular dysfunction, outperforming traditional biomarkers for this condition. This study looked into almost 45,000 patients, which suggests AI could one day help predict the likelihood of heart failure long before the event and/or guide prevention.<sup>2</sup> One important issue to note is that both studies above used historical ECGs rather than real-time patient data, suggesting that these findings will need further validation in prospective studies.

A prospective study tested the ability of AI to detect asymptomatic atrial fibrillation (AF) events in patients with an apparently normal sinus rhythm. In the high-risk group, AI was able to identify AF in 7.6% of flagged individuals, compared to 1.6% in those in the control group. These findings were later confirmed with prolonged monitoring.<sup>3</sup> This approach may allow us to better predict and so prevent complications such as strokes and plan for early interventional therapies for patients that may need it, like ablation. The study, however, had a relatively short period of follow-up (a median of 9.9 months).

These studies show the possibility of growth that AI has in influencing ECG analysis (Table 1), improving accuracy and expanding access to an essential diagnostic tool. While the benefits of AI in ECG analysis are clear, the reliance on retrospective and short-term data shows the need for more prospective studies before AI can be used routinely in clinical cardiology.<sup>1-3</sup>

### AI In Cardiovascular Imaging

Cardiovascular imaging, such as echocardiography and coronary CT, plays an important role in diagnosing heart conditions. However, these imaging modalities are generally complex and require significant expertise for an accurate diagnosis. This is where artificial intelligence has also started making a difference, helping with the fast and accurate interpretation of these otherwise challenging scans.

Echocardiography is widely used to evaluate myocar-

dial contractility and to assess valvular heart functions. AI has shown the ability to interpret these images and video clips efficiently and with a high degree of accuracy; one study found that AI could measure ejection fraction from echo videos with an accuracy comparable to cardiologists.<sup>5</sup> This will reduce the time spent on manual analysis, meaning patients can have their results available in minutes rather than in hours.

Coronary CT angiography is widely used to assess the presence and the severity of coronary artery disease. Here the use of AI has shown good accuracy, with one study reporting AI's ability to detect coronary artery disease with a 90% sensitivity, which matches expert interpretations in most cases.<sup>6</sup> As AI is exposed to thousands of scans, it has the potential to become even faster and more accurate than an average cardiologist. Cardiac MRI is another modality where AI is showing significant promise: it can measure heart volumes and detect scar tissue from previous infarctions. A recent trial found that AI could process MRI images within seconds, instead of minutes, while still maintaining good accuracy.<sup>7</sup> This can be very significant in high-volume clinical settings, or where expert interpretation is otherwise not available.

While these AI tools show potential in making cardiovascular diagnosis more efficient and accessible, most are based on retrospective data. These also often lack sufficient diversity in patient populations and diagnostic findings. This highlights the need for further testing in real-world settings to ensure these methods are truly reliable.

### AI In Risk Prediction

Risk prediction tools such as Framingham and CHA2DS2-VASc Stroke Score used to assist physicians in identifying patients at risk of conditions such as heart attacks or strokes. However, these tools tend to generalize patients based on a limited set of data. AI can help here, by using larger data sets to make more accurate predictions. One study looked into the use of AI to predict heart failure hospitalizations in patients and it found that it had improved accuracy over more traditional methods.<sup>8</sup> The AI model achieved an area under the curve AUC of 0.85, compared to 0.75 for Framingham's.<sup>8</sup> The AI system used trends and data such as changes in blood pressure and lab results, allowing for a more preventive approach, and identifying risks prior to

Table 1: AI applications mode in detecting ECG changes with related references

| AI Application Mode          | Detected Condition           | Potential Clinical Impact                   | Evidence Base                                                | Key Study Limitations                     |
|------------------------------|------------------------------|---------------------------------------------|--------------------------------------------------------------|-------------------------------------------|
| Rhythm analysis              | Atrial Fibrillation (AF)     | Early detection, stroke prevention          | Smart watch integration                                      | Short follow-up (9.9 months) <sup>4</sup> |
| Subtle ECG anomaly detection | Left Ventricular Dysfunction | Predict sudden cardiac death, guide therapy | Deep learning on 12 - lead ECG, ~45,00 patients <sup>3</sup> | Stored ECGs                               |
| Waveform Interpretation      | ST-Segment/QT Prolongation   | Faster, accurate diagnosis                  | Improved sensitivity <sup>1</sup>                            | Retrospective data                        |

further escalation. Another study looked into predicting survival after a heart attack. It used a combination of age, heart function, and blood tests. The study found an improvement of around 10% in predicting survival rates when compared to other scoring systems like the GRACE and TIMI scores.<sup>9</sup> As with other studies, the general limitations of the use of past patient records apply. Here, the dataset is also predominantly older male patients, and so it's unclear how well it may perform in other demographic groups. With AI allowing for a more personalized approach when predicting risk for patients, it can increase accuracy and so improve treatment. However, until these tools are used in real-time patients and for a considerable duration of time, it's hard to fully quantify their clinical value.

## Conclusion

AI holds great potential to improve cardiovascular diagnostics and risk prediction. It can enhance the accuracy of ECG interpretation and improve the efficiency of cardiovascular imaging. It can also provide more personalized risk assessments for patients. AI holds the promise to becoming an essential tool in clinical cardiology in the near future. Over and above the current capabilities to diagnose cardiovascular diseases, the future will see AI being integrated into more personalized therapeutic and diagnostic decisions. However, despite its potential, AI, it introduces a new level of data protection, ethical, and legal challenges in modern health care.<sup>10, 11</sup>

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